## Task

Analyze the socio-economic indicators of countries from the "Country-data.csv" dataset to identify distinct country segments. This analysis should include data preprocessing (outlier handling, scaling), dimensionality reduction using PCA, and a comparison of clustering techniques (K-Means, Hierarchical, DBSCAN). Finally, interpret the results to propose actionable policy recommendations and provide an executive summary of the findings.

## Define Problem & Load Data

Subtask:
Outline the problem statement, objectives, and the process of loading and initially inspecting the `Country-data.csv` dataset, including checking data types, summary statistics, and missing values.

## Data Preprocessing & EDA

Detail the steps for handling outliers (e.g., using IQR capping), visualizing data distributions (histograms, boxplots), analyzing skewness, and exploring feature relationships through correlation heatmaps and pairplots. Also, include identifying top/bottom countries for key indicators.

## Feature Scaling

Subtask:
Standardize numerical features using StandardScaler.
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## Dimensionality Reduction (PCA)

Subtask:
Describe the application of Principal Component Analysis (PCA) to the scaled data, including selecting the optimal number of components to retain sufficient variance. Discuss interpreting the principal components through their loadings and visualizing the data in the PCA space (e.g., PC1 vs PC2).

## Clustering with K-Means

Explain the K-Means clustering process, including how to determine the optimal number of clusters (k) using methods like the Elbow method and Silhouette score. Detail how to apply K-Means to the PCA-reduced data, visualize the resulting clusters, profile each cluster based on original feature means, and identify representative countries.

## Clustering with Hierarchical Method

Subtask:
Apply Hierarchical Clustering, visualize the dendrogram, cut it to form clusters, visualize these clusters in PCA space, and compare the results with K-Means.

## Clustering with DBSCAN

Subtask:
Detail the DBSCAN clustering approach, focusing on how to determine optimal parameters like `eps` using k-distance plots. Explain how to apply DBSCAN, visualize its clusters and identified noise points in the PCA space, and list countries identified as noise.

## Evaluate & Compare Clustering Models

Subtask:
Outline the process of evaluating and comparing the different clustering algorithms (K-Means, Hierarchical, DBSCAN) using internal validation metrics such as Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index.

## Interpret Results & Propose Recommendations

Subtask:
Summarize the key findings from the clustering analysis, specifically focusing on the profiles of the identified country segments. Provide actionable business interpretations and policy recommendations based on these segments. Discuss potential next steps for further analysis.